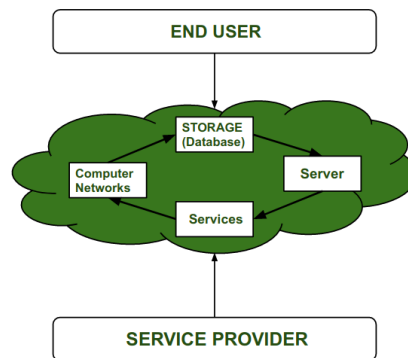


Chapter 1: Introduction to Cloud Computing

Origins of Cloud computing:



Era 1: The Mainframe Era (1950s – 1960s)

- The Problem: Computers were the size of entire rooms and cost millions. Only giant corporations or governments could own one.
- The Solution: Mainframes. Multiple users connected to one "beast" of a machine using "dumb terminals" (screens with no processing power).
- Key Concept: Time-Sharing. This was the "Grandfather of the Cloud." It allowed multiple people to use the same CPU at once, so the machine was never sitting idle.

Era 2: The Birth of the "Utility" Idea (1961)

- The Visionary: John McCarthy (who also coined the term "Artificial Intelligence") famously said at MIT: "Computing may someday be organized as a public utility just as the telephone system."
- The Meaning: He predicted that one day, you wouldn't buy a computer; you would just "plug in" to the wall for processing power and pay for what you use exactly like your water or electricity bill today.

Era 3: Client-Server & Distributed Computing (1980s – 1990s)

- The Shift: Computers became small enough for everyone to have one on their desk (PCs).
- The Model: Instead of one giant brain, we had Clients (your PC) requesting data from Servers (the office computer).
- Distributed Computing: To solve massive problems, scientists began linking thousands of these computers together over a network to act as one.

Era 4: The SaaS Revolution (1999)

- The Breakthrough: Salesforce.com changed the game. They were the first to deliver a "business app" (CRM) entirely through a website.
- Why it mattered: For the first time, a company didn't have to install software on 500 laptops. They just opened a browser. This proved that "Software as a Service" (SaaS) was the future.

Era 5: The Modern Cloud Giants (2002 – Present)

- 2002: Amazon realized they had massive amounts of "extra" server space when it wasn't the holiday shopping season. They decided to rent that space to other companies. This became AWS.
- 2006: AWS launched EC2 (Elastic Compute Cloud), allowing anyone to rent a virtual computer by the hour.
- 2008-2010: Google App Engine and Microsoft Azure launched, turning the cloud into a global competition.

Essential characteristics:

1. **On-demand self-services:** The Cloud computing services do not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.
2. **Broad network access:** The Computing services are generally provided over standard networks and heterogeneous devices.
3. **Rapid elasticity:** The Computing services should have IT resources that are able to scale out and in quickly and on a need basis. Whenever the user require services it is provided to him and it is scale out as soon as its requirement gets over.
4. **Resource pooling:** The IT resource (e.g., networks, servers, storage, applications, and services) present are shared across multiple applications and occupant in an uncommitted manner. Multiple clients are provided service from a same physical resource.
5. **Measured service:** The resource utilization is tracked for each application and occupant, it will provide both the user and the resource provider with an account of what has been used. This is done for various reasons like monitoring billing and effective use of resource.
6. **Location-Independent Resource Pooling in Cloud Computing**
Location-independent resource pooling means that cloud computing resources such as servers, storage, memory, processing power, and databases are pooled together by the cloud provider and made available to users without requiring users to know the exact physical location of the resources.

How it works

Cloud providers maintain large data centers containing many physical and virtual resources.

Through virtualization and resource management, these resources are dynamically allocated to users according to their requirements.

For example, when a student uploads a dataset to a cloud storage service, the student does not need to know whether the data is physically stored on a server in Mumbai, Pune, Singapore, or another data center. The cloud service manages the underlying resources.

Roots of Cloud Computing

The **roots of cloud computing** refer to the technologies and concepts that gradually evolved into modern cloud computing. Cloud computing did not appear suddenly; it developed from several earlier computing models.

1. Mainframe Computing

In the **1960s and 1970s**, organizations used powerful mainframe computers that were shared by many users through terminals.

Key idea: Centralized computing resources can be shared by multiple users.

Example: A university could have one powerful mainframe accessed by many students.

2. Time-Sharing

Time-sharing allowed multiple users to use the same computer simultaneously. The computer divided its processing time among different users.

Key idea: Efficient sharing of computing resources.

This concept is similar to today's **multi-user cloud environments**.

3. Client-Server Computing

In the **1980s and 1990s**, computing systems increasingly used a client-server model.

- **Client:** Requests services or data.
- **Server:** Provides processing, storage, or applications.

Key idea: Computing resources and services can be provided remotely over a network.

4. Distributed Computing

Distributed computing involves multiple computers working together to perform a task.

For example, a large data-processing task can be divided among several computers.

Key idea: Multiple computers can work together as a single computing environment.

5. Internet

The growth of the **Internet** made it possible to access computing services from different geographical locations.

Instead of installing applications and storing all data locally, users could access resources through the Internet.

Key idea: Computing services can be delivered over a network.

6. Virtualization

Virtualization became one of the most important foundations of cloud computing.

A single physical server can be divided into multiple **virtual machines (VMs)**.

For example:

One physical server → **VM 1 + VM 2 + VM 3 + VM 4**

This improves resource utilization and allows cloud providers to allocate resources dynamically.

